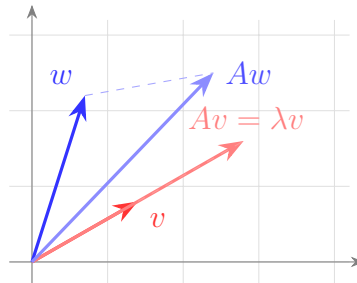


Explanation – HW11: Eigenvalues, Eigenvectors & Eigenspaces

Part 1: Foundations – explained simply

1. What is an eigenvector? “The direction that stays”

A matrix A is a **deformation of space**: it takes a vector and makes a new one out of it. Usually the vector is *rotated and stretched* in the process. But there are special directions that are **only stretched** during the deformation – afterwards they point exactly where they did before. These directions are the **eigenvectors**, and the stretch factor is the **eigenvalue** λ .



Analogy

Imagine a **dough** that you pull apart. A little arrow drawn on it at a slant usually rotates as you pull. But an arrow pointing exactly in the pulling direction keeps its direction – it only gets longer. That is exactly an eigenvector, and “how much longer” is the eigenvalue.

Remember

$v \neq 0$ is an **eigenvector** of A for the **eigenvalue** λ if

$$Av = \lambda v.$$

The zero vector never counts as an eigenvector (otherwise everything would be trivial).

2. How do you find the eigenvalues? The characteristic polynomial

We rewrite $Av = \lambda v$: $Av - \lambda v = 0$, i.e. $(A - \lambda I)v = 0$. We are looking for a $v \neq 0$ that solves this equation. But a homogeneous system has a solution $\neq 0$ *only if* the matrix $A - \lambda I$ is **not invertible** – and that means $\det(A - \lambda I) = 0$.

The key

$$\lambda \text{ is an eigenvalue} \iff \det(A - \lambda I) = 0.$$

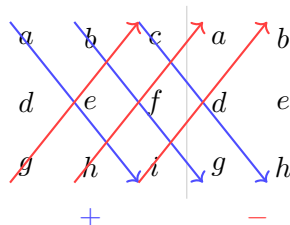
You compute $\chi_A(t) = \det(A - tI)$ (the **characteristic polynomial**) and look for its roots.

Analogy

$\det$ measures whether a matrix “squashes” directions flat. $\det(A - \lambda I) = 0$ means: $A - \lambda I$ squashes at least one direction down to 0. Exactly that direction v survives as an eigenvector, because $(A - \lambda I)v = 0$ means $Av = \lambda v$.

3. Tool A: The Sarrus rule (determinant of a 3×3)

For a 3×3 matrix you compute the determinant with Sarrus: write the first two columns again on the right, multiply the three diagonals going *down* (+) and subtract the three diagonals going *up* (–).



Sarrus formula

$$\det = \underbrace{aei + bfg + cdh}_{\text{downward (+)}} - \underbrace{ceg + afh + bdi}_{\text{upward (-)}}.$$

4. Tool B: Guessing integer roots

Once the polynomial is multiplied out (e.g. $t^3 - 11t^2 + 39t - 45$), you have to factor it. Trick: **integer roots are always divisors of the constant term**. For -45 that means $\pm 1, \pm 3, \pm 5, \pm 9, \pm 15, \pm 45$. You substitute them one after another until 0 comes out.

Two free checks

- Sum of all eigenvalues = $\text{trace}(A)$ (sum of the diagonal).
- Product of all eigenvalues = $\det(A)$.

This often gives you the last root without computing.

5. Tool C: Factor out early (the elegant way)

Instead of multiplying everything out first, you can often *simplify* $\det(A - tI)$ before computing. If, for instance, **every column of A has the same sum s** , then in $A - tI$ every column sums to $s - t$. If you add all rows into the first ($R_1 \rightarrow R_1 + R_2 + \dots$), the first row becomes all $(s - t)$ – and $(s - t)$ can be pulled out as a factor. This immediately gives an eigenvalue $\lambda = s$ and makes the rest small.

Analogy

This is like cancelling *before* multiplying out: you don't compute $\frac{6 \cdot 7}{6}$ either, you cancel the 6 first. In the same way you pull out the factor $(s - t)$ before the computation gets big.

6. Eigenspace, AM and GM

Once you have an eigenvalue λ , you solve $(A - \lambda I)v = 0$ with Gauss. All solutions together form the **eigenspace** E_λ .

Two multiplicities – don't mix them up!

- **AM**(λ) (algebraic): how often $(t - \lambda)$ sits in the char. polynomial. (*From the polynomial.*)
- **GM**(λ) (geometric): $\dim E_\lambda$ = number of free variables when solving $(A - \lambda I)v = 0$. (*From the Gauss.*)
- It always holds that $1 \leq \text{GM}(\lambda) \leq \text{AM}(\lambda)$.

how often a root

dim eigenspace

AM = 2

$\leq$

GM

If *all* GM = AM,
 A is diagonalizable.

Analogy

AM is **how many seats** an eigenvalue has “reserved”; GM is **how many it actually fills with its own directions**. Filling more than reserved is impossible ($\text{GM} \leq \text{AM}$). Only if every eigenvalue fills all its seats do you have enough eigenvectors for a whole basis – then A is **diagonalizable**.

7. Tool D: Eigenvalues without a polynomial (for Exercise 44)

Sometimes you can read eigenvalues off directly, without computing:

Useful propositions

- **A not invertible** ($\det A = 0$) $\Leftrightarrow 0$ is an eigenvalue. ($\text{GM}(0) = \dim \ker A$.)
- If a column j appears as $\lambda \cdot e_j$ (just one entry λ on the diagonal, rest 0), then e_j is an eigenvector for λ .
- **Rank-1 matrix** (all rows multiples of the same row): the eigenvalues are trace (once) and 0 (otherwise).
- Sum of eigenvalues = trace, product = det – use this to fill gaps.

Part 2: Overview of the exercises

Exercise 43 – char. polynomial in two ways

Two 3×3 matrices, the full program each. The exercise explicitly demands **both ways**:

- **Method 1:** Sarrus $\rightarrow$ multiplied-out polynomial $\rightarrow$ guess integer roots (divisors of the constant term) $\rightarrow$ factor.
- **Method 2:** column sums are constant (5 for A , 4 for B) $\Rightarrow$ pull out the factor $(s - t)$ early, the rest to triangular form with column operations.

Then solve $(A - \lambda I)v = 0$ per eigenvalue. **Punchline:** A is diagonalizable (all GM = AM), B is *not* – for B the eigenvalue $\lambda = 2$ has AM 2 but only GM 1.

Exercise 44 – eigenvalues without a polynomial

Here the char. polynomial is **forbidden**; you argue with properties.

- A : row 1 = row 3 $\Rightarrow$ singular $\Rightarrow \lambda = 0$. Middle column $(0, 6, 0) \Rightarrow e_2$ is an eigenvector for 6. The rank of $A - 6I$ is 1, so GM(6) = 2; the trace 12 confirms $0 + 6 + 6$.
- B : all rows multiples of $(1, 1, 1, 1) \Rightarrow$ rank 1 $\Rightarrow$ GM(0) = 3. Trace = $5 + 7 + 11 - 23 = 0 \Rightarrow$ there is no further eigenvalue $\Rightarrow \lambda = 0$ with AM 4. Because GM 3 < AM 4, it is *not* diagonalizable.

Exercise 45 – four short proofs

Always the same pattern: take $v \neq 0$ with $Av = \lambda v$ and check that v is also an eigenvector of the new matrix.

- (a) $(A + cI)v = \lambda v + cv = (\lambda + c)v$.
- (b) $A^2v = A(\lambda v) = \lambda^2 v$.
- (c) For invertible A , $\lambda \neq 0$; from $Av = \lambda v$ it follows that $A^{-1}v = \frac{1}{\lambda}v$.
- (d) $(AB)v = A(\mu v) = \lambda \mu v$ (if $Av = \lambda v$, $Bv = \mu v$).

Summary

Matrix	Eigenvalues (AM, GM)	diagonalizable?
43 A	3 (2, 2), 5 (1, 1)	yes
43 B	2 (2, 1), 4 (1, 1)	no
44 A	0 (1, 1), 6 (2, 2)	yes
44 B	0 (4, 3)	no

Golden rules

1. Eigenvalues = roots of $\det(A - tI)$.
2. Trace = sum of eigenvalues, det = product – always use as a check.
3. Constant column sums $s \Rightarrow$ factor $(s - t)$ and eigenvalue s for free.
4. $\det = 0 \Leftrightarrow 0$ is an eigenvalue; rank-1 matrix $\Rightarrow$ eigenvalues trace and 0.
5. Per eigenvalue: solve $(A - \lambda I)v = 0 \rightarrow$ basis $\rightarrow$ GM = number of free variables.

6. All $\text{GM} = \text{AM} \Rightarrow$ diagonalizable.